

Bin-Picking Project Initial Evaluation Form



This evaluation form must be filled out in order for Scape Technologies to evaluate the feasibility of a bin-picking project. Further information is available at the end of this document.

General Project and Work Cell Information				
1.01	Project Name			
1.02	Total of different parts in project			
1.03	Bin type (EU-Pallet, Metal solid, etc.) Include picture			
1.04	Bin Inner Dimensions [mm]			
1.05	Preferred Robot Brand and model (if any)			
Part Information (1 st Part in Project)				
2.01	Part Name (or Product Number)			
2.02	Part Dimensions [mm] (WxBxH)			
2.03	Part Weight [kg] and Material			
2.04	Desired Average Cycle Time [sec]		2.05	Average based on [Cycles]
2.06	CAD file available for the part? See description for preferred formats.	☐ Yes ☐ No	2.07	Any oil/soap on the part? ☐ Yes ☐ No
2.08	Are the parts separated by a slip sheet?	☐ Yes ☐ No	2.09	Can the parts be entangled? ☐ Yes ☐ No
2.10	Any temperature issues?	☐ Yes ☐ No	2.11	Is the part very shiny? ☐ Yes ☐ No
2.12	Short description of place requirements:			
2.13	Determine which side is up?	☐ Yes ☐ No	2.14	Any special gripper requirements? ☐ Yes ☐ No
Part Information (2 nd Part in Project – if available)				
2.01	Part Name (or Product Number)			
2.02	Part Dimensions [mm] (WxBxH)			
2.03	Part Weight [kg] and Material		0.917 weight of two items together	
2.04	Desired Average Cycle Time [sec]		2.05	Average based on [Cycles]
2.06	CAD file available for the part? See description for preferred formats.	☐ Yes ☐ No	2.07	Any oil/soap on the part? ☐ Yes ☐ No
2.08	Are the parts separated by a slip sheet?	☐ Yes ☐ No	2.09	Can the parts be entangled? ☐ Yes ☐ No
2.10	Any temperature issues?	☐ Yes ☐ No	2.11	Is the part very shiny? ☐ Yes ☐ No
2.12	Short description of place requirements: Our company is engaged in automating the welding process of metal parts. We need a system to recognize and retrieve the parts that the robot will deliver to the welding table. The prevailing temperature during the welding process does not concern the planned robot, as it only unloads the parts on the table. (Another robot takes the finished parts)			

2.13	Determine which side is up?	☐ Yes ☐ No	2.14	Any special gripper requirements?	☐ Yes ☐ No
Form Filled by:					
Company		Telephone			
Name		E-mail			
Date					
Things to collect:					
	Part CAD file (3D)		Bin/pallet image(s) with parts in it		
	Physical parts (Quantity to be decided by Scape)		Images of where the parts must be placed/delivered		

Description

Items written with bold are most important. This sheet can hold information about two different parts. Please use additional sheets if there are more parts.

Item	Description
1.01	Name of the project which we can refer to when communicating with you.
1.02	The total number of different parts in the project.
1.03	The bin-type can be an EU-Pallet (half or full), Metal (solid or lattice), plastic box (color), etc. Please describe the bottom (if not available on picture). Is it flat, lattice, or <i>wavy</i> ?
1.04	The inner dimensions of the bin are important in order to select a robot which can reach all corners.
1.05	If a specific robot brand and model is preferred please let us know.
2.01	The name of the part and/or a product number which uniquely defines the part.
2.02	The approximate dimensions of the part measured in mm. This is just to give us an idea about the size.
2.03	The weight of the part. This is important regarding the choice of gripper.
2.04	The desired average cycle time is crucial information. It is defined as the time between the delivery of one part to its final destination until the delivery of the next part to its final destination.
2.05	The cycle time in 2.04 is always based on an average of a number of picked parts. This information is important because cycle times for bin-picking vary. Instead of writing a number of cycles it is also possible to write for instance: "1 bin" or "1 shift".
2.06	Check yes, if you have a 3D CAD file of the part. The format needed by the SCAPE system is STL and typically about 0.01 mm accuracy. Other formats such as STP/STEP, IGS/IGES, or DWG/DXF can be converted by Scape.
2.07	If there is any oil or soap or other lubrication on the parts please state it here.
2.08	Sometimes parts are in layers separated by sheets. In this case a sheet remover tool must be built for the robot tool-unit. It is typically a frame with several suction cups that can be picked up automatically by the robot.
2.09	If there is any risk of entanglement (or in any other way stick together) please state it here. In this case, Scape will probably need more parts for testing this risk.
2.10	If the environment or the parts are very hot, this can influence the type of gripper. Add a note at the bottom about expected temperature if above 50 degrees.
2.11	Parts are considered very shiny when they can reflect/mirror neighboring parts. This can for instance be polished metal parts.
2.12	Please give a short description of the place requirements. This can be a fixture or a machine in which the parts must be placed. Include images if possible. If you know the tolerances, they can also be mentioned here. Scape will in most cases discuss the place requirements at some point because it is crucial to understand this correct.
2.13	In most bin-picking projects, it is necessary to make a precision grip. This is done by use of the SCAPE Handling Station. Essentially the Handling Station is just a table on which a single part is placed and re-gripped. Sometimes it is important to re-orient the part in the gripper after the Handling Station because it cannot be delivered to its final place. For instance, if the side we need to grip on is facing downwards on the Handling Station, then we need an extra re-grip afterwards. In this case check the Yes box. Another similar example is flat parts on which there is a small text on one side but not on the other. Often SCAPE cannot distinguish between these two cases. If it is necessary to distinguish between these two cases, please check the YES box.
2.14	Sometimes there are special gripper requirements. For instance a special gripper must be available on the robot tool-unit in order to place the part in its final destination or to hold the part during a certain task. Check the YES box if this is the case. Scape will then make sure that there is room on the tool-unit for the gripper.

Notes: